

CWC Generator

Prototype One

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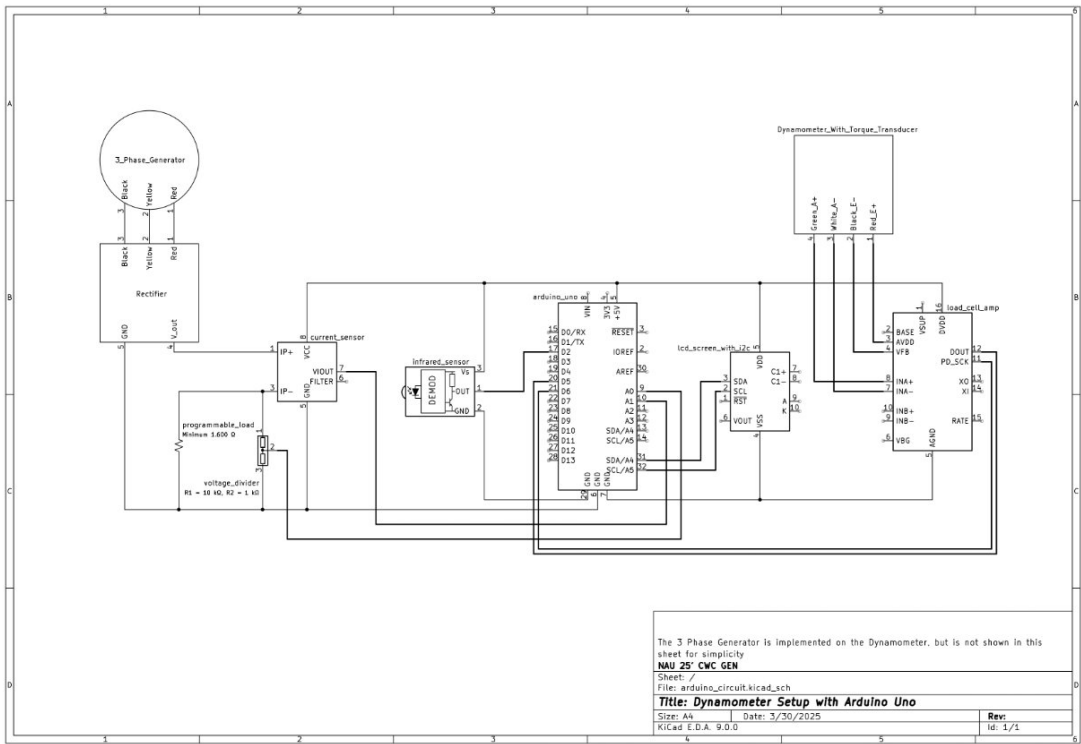
MOTOR-CAD



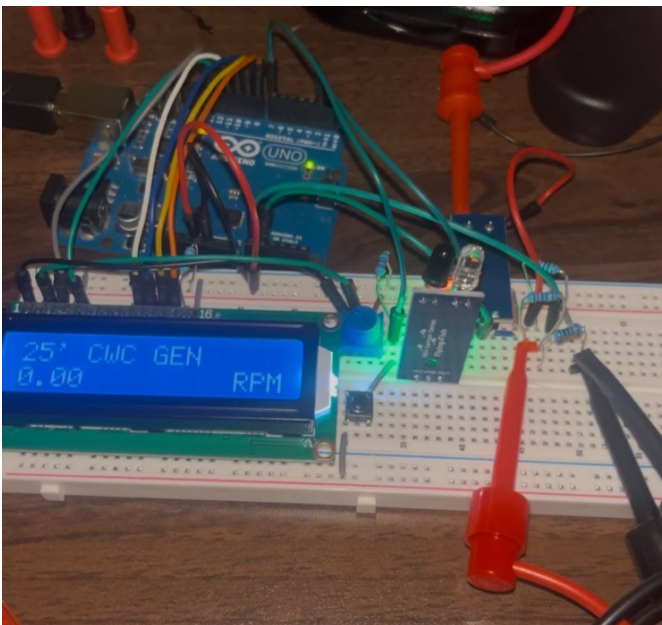
MAXWELL

Physical Prototype Dynamometer

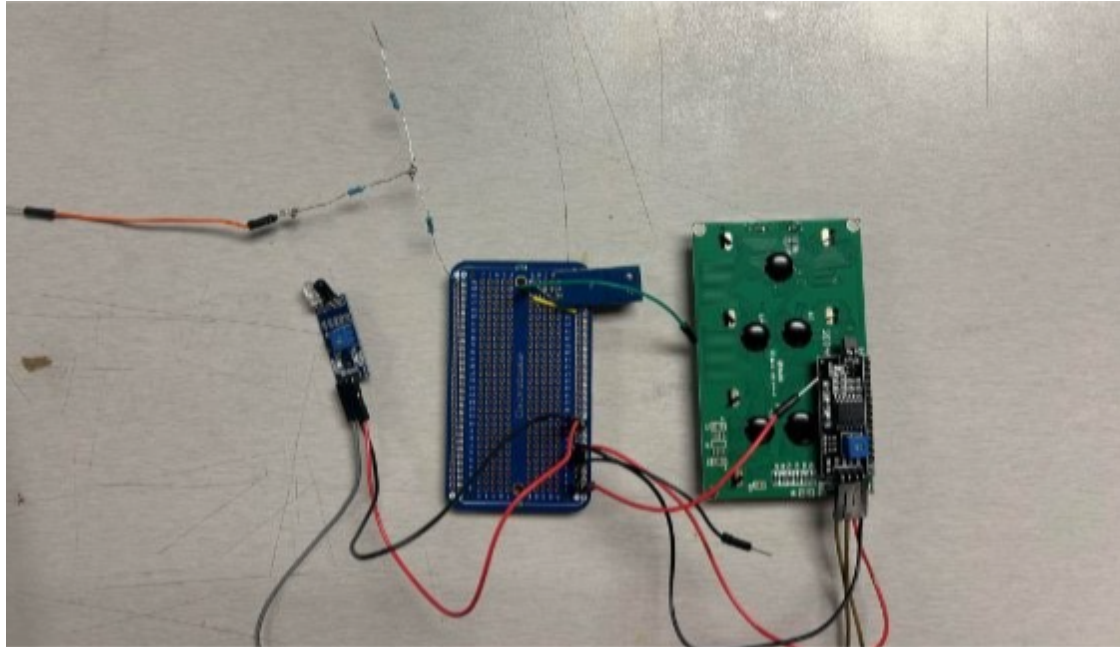
Final Schematic



Arduino Circuit in Breadboard



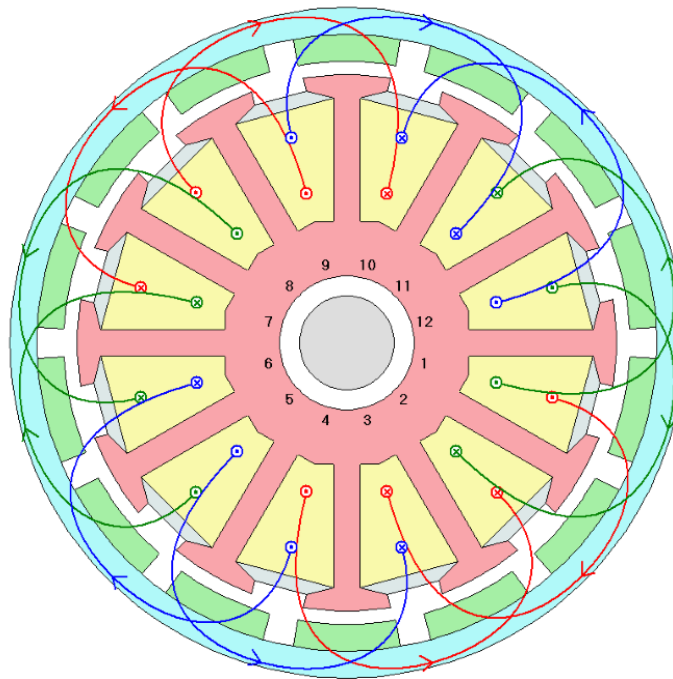
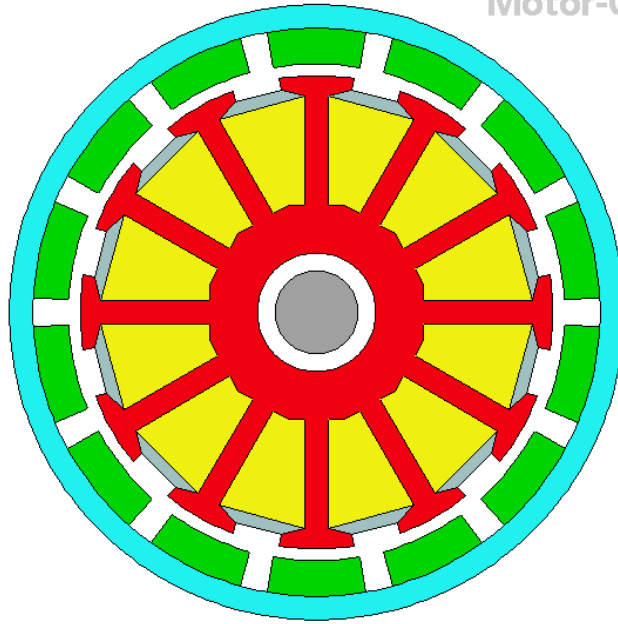
Permanent Arduino Circuit



The question we are trying to answer is how can we make the Dynamometer a precise test bench? The answer to the question is giving the Dynamometer intelligence with an Arduino that will take inputs of voltages and signals that are converted to torque, rotations per minute (RPM), current, and voltage. With these conversions, it'll go through an algorithm that will refine the values to an accurate measurement and display them on a liquid crystal display (LCD). Therefore, we get a Dyno that gives out precise measurements. We finalized this Dyno design by following its Quality Function Deployment customer requirements of safety, since the maximum allowable amperes is 30 A, user-friendliness, and preciseness of all measurements.

Modeling Number of Turns in a Coil in MotorCAD, Motor Melon

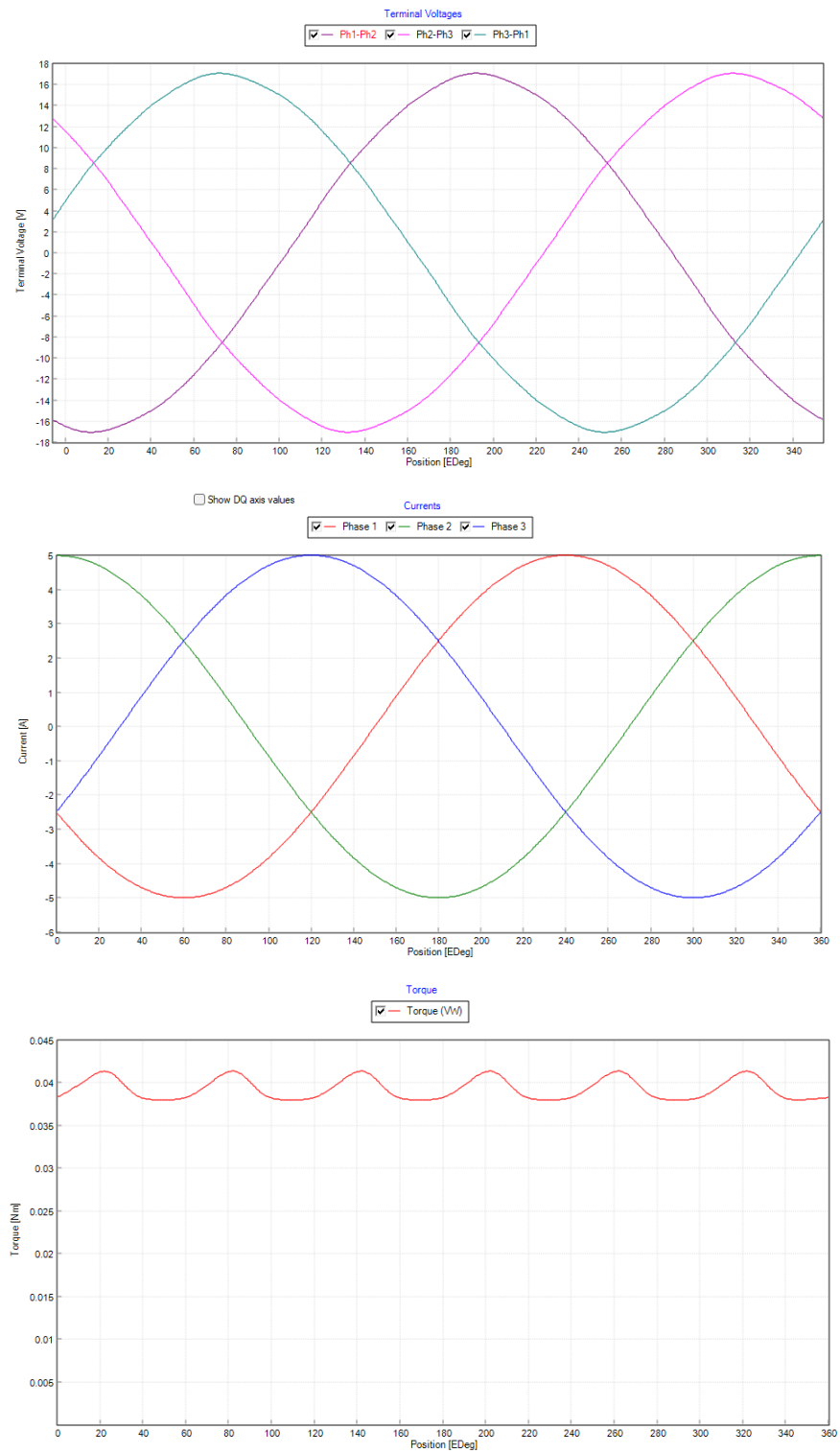
3D Model



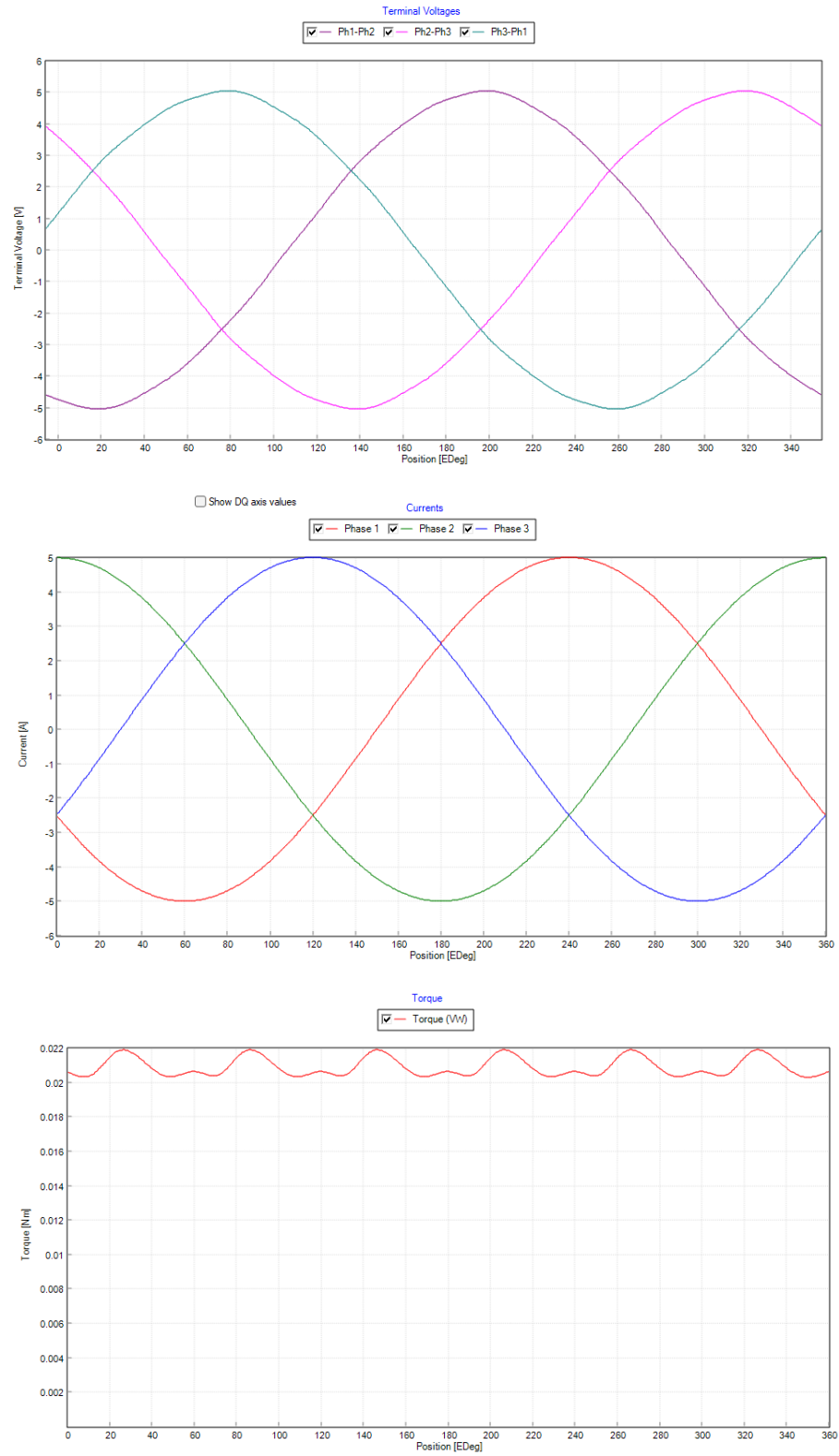
Key:

Green-	Blue-	Yellow-	Red-	Grey-
Magnets	Case	Coils	Stator	Shaft

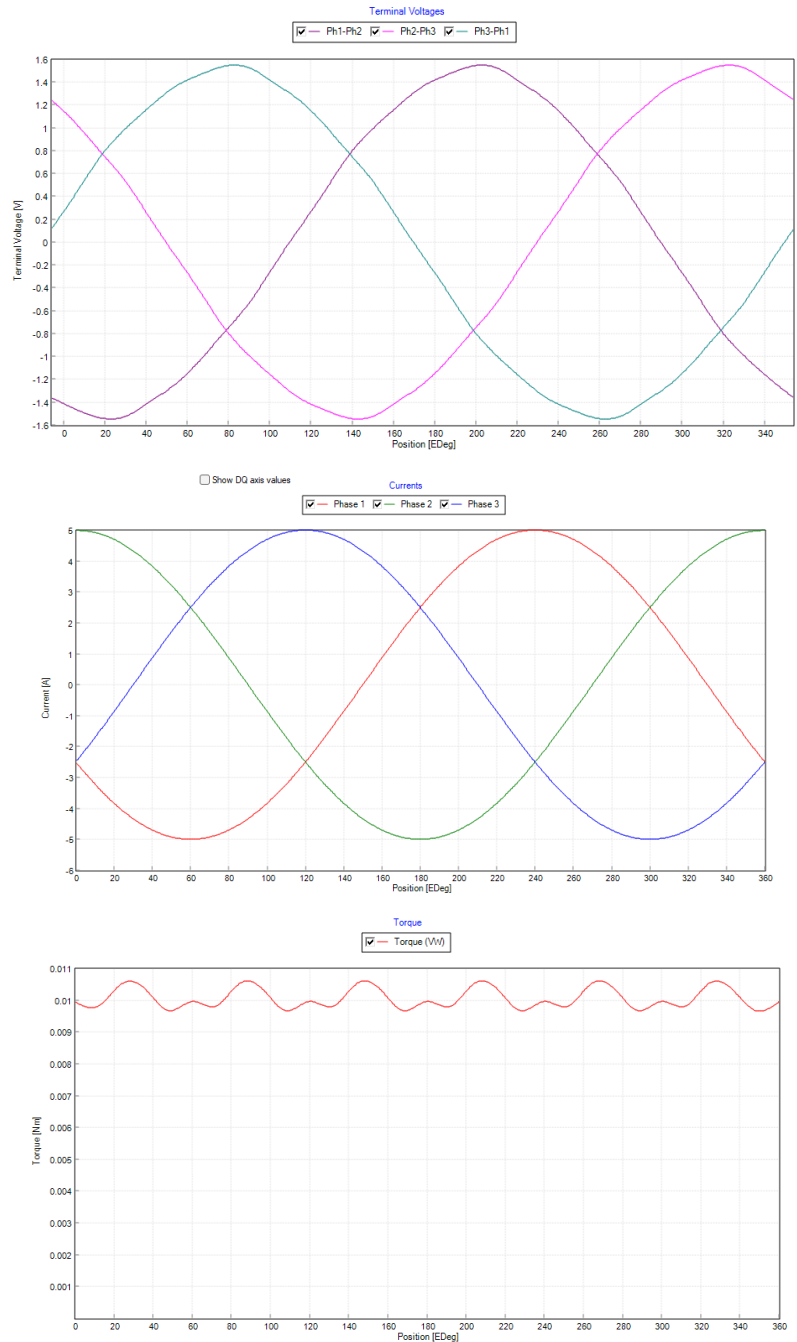
50 Turns in a Coil



25 Turns in a Coil



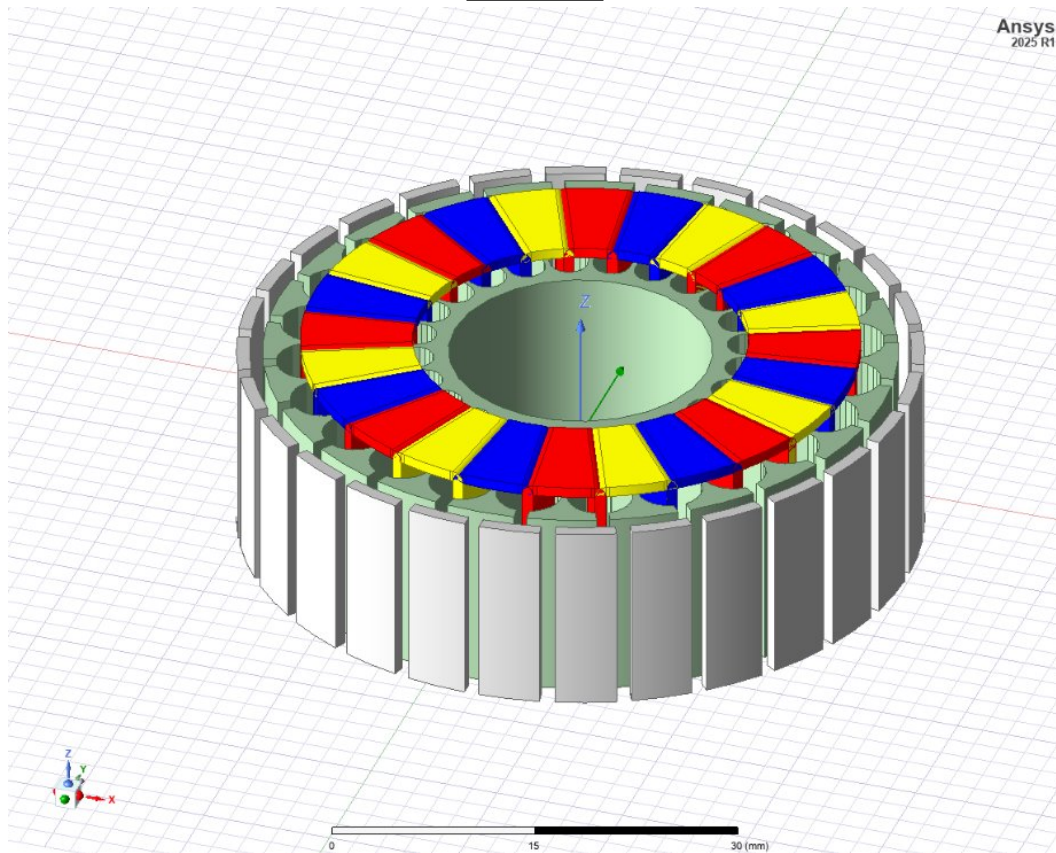
12 Turns in a Coil



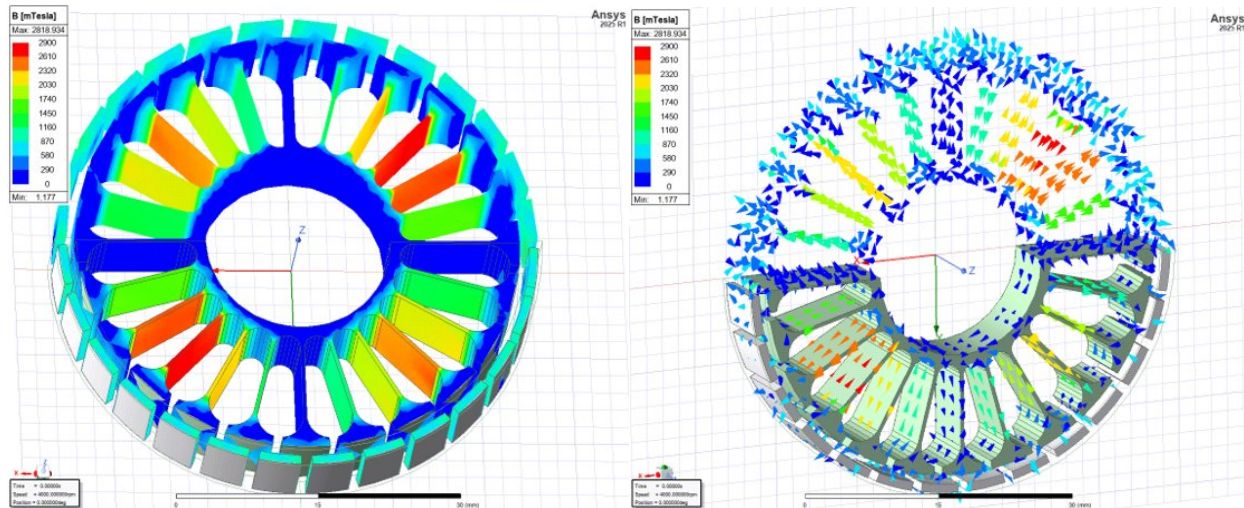
Three different simulations were run with three different coil variations, 50, 25, and 12 turns. From these simulations we learned that the higher number of turns produces a higher amount of torque and voltage. These simulations will be used to inform our future designs because they have proven the effects of the turn count.

Modeling MADJENNY in Ansys Maxwell

3D Model



Surface and vector simulations of magnetic flux



What these simulations tell us is how much magnetic flux is being induced by the generator at any given time. The Tesla induced can be used to tell us induced current and voltage depending on parameters of the generator, like wire length and gage.